

CS 105: Department Introductory Course on Discrete Structures

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Aug 22, 2025

Lecture 07 – Basic structures: sets and functions

Logistics

Quiz 1: Aug 28th, 3.45pm

Venue: LH classrooms (see Piazza next week for precise seating arrangement!)

Optional Problem Solving Sessions started...

- ▶ 3.30pm Wednesdays (Same venue: CC101/103/105)
- ▶ Tomorrow: We may have a special session for basic problems (see piazza)

Safe and Attendance

- ▶ Any remaining issues: Private message on Piazza to Instructors with proof (screenshots etc).

From proofs to structures

Chapter 2: Basic Discrete Structures

- ▶ Finite sets and Russell's Paradox
- ▶ Infinite Sets and comparing them using functions

Hilbert's hotel



- ▶ Suppose there is a hotel with infinitely many rooms.
 - ▶ And suppose they are all full (like in IIT guest house).
1. Can you accomodate 1 or finitely many more guests, by shifting around the existing guests?
 2. What if infinitely many more guests arrive?
 3. What if infinitely many more trains with infinitely many more guests arrive?
- (H.W)

Difference between finite and infinite sets

Theorem

Let A be a set and $b \notin A$. A is infinite iff there is a bijection from A to $A \cup \{b\}$.

1. If A infinite, then $A \neq \emptyset$, so let $a_0 \in A$. Define $f(a_0) = b$.
2. Now $A \setminus \{a_0\}$ is infinite $\implies$ non-empty, so let $a_1 \in A \setminus \{a_0\}$. Define $f(a_1) = a_0$.
3. $\forall i \in \mathbb{N}, i \geq 1$, $A \setminus \{a_0, \dots, a_{i-1}\}$ is infinite, hence non-empty, so let $a_i \in A \setminus \{a_0, \dots, a_{i-1}\}$. Define $f(a_i) = a_{i-1}$.
4. Collecting all such a'_i 's, we get a subset $A' = \{a_i \in A \mid i \in \mathbb{N}\} \subseteq A$. (Note it may be that $A \neq A'$).
5. Now, $\forall a \in A$ if $a \notin A'$, we define $f(a) = a$.
6. f is a bijection! Prove surjection and injection...

Corollary: Difference between finite vs infinite sets

- Even if A, B are infinite, $A \subsetneq B$, there can be a bijection from A to B , i.e., they have the same “cardinality”.
- From any set A , there is a surjection from A to $\mathbb{N}$.

Comparing infinite sets using functions

Theorem

There is a bijection from $\mathbb{Z}$ to $\mathbb{N}$.

Proof: Hilbert's hotel argument. But how do you formalize it?

$$f(x) = \begin{cases} -2x & \text{if } x \leq 0 \\ 2x - 1 & \text{else} \end{cases}$$

Some questions...

1. Is there a bijection between $\mathbb{N} \times \mathbb{N}$ to $\mathbb{N}$?
2. Is there a bijection between $\mathbb{N} \times \mathbb{N} \times \mathbb{N}$ to $\mathbb{N}$?
3. Is there a bijection from $\mathbb{Q}$ to $\mathbb{N}$?
4. Is there a bijection from the set of all subsets of $\mathbb{N}$ to $\mathbb{N}$?
5. Is there a bijection from $\mathbb{R}$ to $\mathbb{N}$?

Countable and countably infinite sets

Definition

- ▶ For a given set C , if there is a bijection from C to $\mathbb{N}$, then C is called **countably infinite**.
- ▶ A set is **countable** if it is finite or countably infinite.

Examples: even numbers, number of horses,...

By previous corollary ($\exists$ surj from any infinite set to $\mathbb{N}$)

Countably infinite sets are the “smallest” infinite sets.

What are the other properties of countable sets?

Some questions...

Are the following sets countable?

That is, is there a bijection from these sets to $\mathbb{N}$?

- ▶ the set of all integers $\mathbb{Z}$
- ▶ $\mathbb{N} \times \mathbb{N}$
- ▶ $\mathbb{N} \times \mathbb{N} \times \mathbb{N}$
- ▶ the set of rationals $\mathbb{Q}$
- ▶ the set of all (finite and infinite) subsets of $\mathbb{N}$
- ▶ the set of all real numbers $\mathbb{R}$

To show these it suffices to show that

- ▶ there is an **injection** from these sets to $\mathbb{N}$
- ▶ or there is a **surjection** from $\mathbb{N}$ (or **any countable set**) to these sets.

Unions of countable sets is countable

Let $A = \{a_0, \dots\}$ be a countably infinite set and B be a set. Then, is $A \cup B$ countable, under the following conditions?

1. $B = \{b_0\}$ is a singleton
2. $B = \{b_0, \dots, b_n\}$ is a finite set
3. $B = \{b_0, \dots\}$ is a countably infinite set
Can we say $\{a_0, \dots, b_0, \dots\}$ is a countably infinite set?

- But then what is the position of b_i (i.e., natural number corresponding to it)?
- Rather, choose $\{a_0, b_0, a_1, b_1, \dots\}$, then b_i is at $(2i + 1)^{th}$ position.
- Formally, define a bijection $f : (A \cup B) \rightarrow \mathbb{N}$ by $f(a_i) = 2i$ and $f(b_i) = 2i + 1$
- Is this correct?

Products of countable sets are countable

Thm: Cartesian product of two countably infinite sets is countably infinite

Proof: Let A, B be countably infinite. Find a way to “number” the elements in $A \times B = \{(a, b) \mid a \in A, b \in B\}$.

► That is, define a bijection from $A \times B$ to $\mathbb{N}$.

$$f(a_i, b_j) = \left(\sum_{k=1}^{i+j} k \right) + j + 1$$

Corollaries

- $\mathbb{N} \times \mathbb{N}$, $\mathbb{N} \times \mathbb{N} \times \mathbb{N}$, $\mathbb{N} \times \mathbb{Z} \times \mathbb{N}$ are countable.
- The set of (positive) rationals is countable.

Hint: Show that $f(a, b) = \begin{cases} a/b & \text{if } b \neq 0 \\ 0 & \text{if } b = 0 \end{cases}$ is a surjection. How does result follow?

Countable sets and functions

Are the following sets countable?

- ▶ the set of all integers $\mathbb{Z}$
- ▶ $\mathbb{N} \times \mathbb{N}$
- ▶ $\mathbb{N} \times \mathbb{N} \times \mathbb{N}$
- ▶ the set of rationals $\mathbb{Q}$
- ▶ the set of all (finite and infinite) subsets of $\mathbb{N}$
- ▶ the set of all real numbers $\mathbb{R}$

Are all sets countable?